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Phonological level wav2vec2-based Mispronunciation Detection and Diagnosis method

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ABSTRACT

The automatic identification and analysis of pronunciation errors, known as Mispronunciation Detection and Diagnosis (MDD) plays a crucial role in Computer Aided Pronunciation Learning (CAPL) tools such as Second-Language (L2) learning or speech therapy applications. Existing MDD methods relying on analysing phonemes can only detect categorical errors of phonemes that have an adequate amount of training data to be modelled. Due to the unpredictable nature of pronunciation errors made by non-native or disordered speakers and the scarcity of training datasets, it is unfeasible to model all types of mispronunciations. Moreover, phoneme-level MDD approaches can provide only limited diagnostic information about the error made. To address this, in this paper, we propose a low-level MDD approach based on the detection of phonological features. Phonological features break down phoneme production into elementary components that are directly related to the articulatory system leading to more formative feedback for the learner. We further propose a multi-label variant of the Connectionist Temporal Classification (CTC) approach to jointly model the non-mutually exclusive phonological features using a single model. The pre-trained wav2vec2 model was employed as a core model for the phonological feature detector. The proposed method was applied to L2 speech corpora collected from English learners from different native languages. The proposed phonological level MDD method was further compared to the traditional phoneme-level MDD and achieved a significantly lower False Acceptance Rate (FAR), False Rejection Rate (FRR), and Diagnostic Error Rate (DER) over all phonological features compared to the phoneme-level equivalent.

1. Introduction

Unlike acoustic modelling for Automatic Speech Recognition (ASR) applications where an abstract model that can handle all variations of the same word is desirable, pronunciation assessment applications such as second language (L2) learning, speech therapy, and language proficiency tests, require accurate detection of any deviation from standard pronunciation.

Mispronunciation detection and diagnosis (MDD) aims to automatically detect pronunciation errors in speech productions and diagnose them by providing error details such as the error location and type as well as a pronunciation quality score. MDD is a key feature of Computer-Aided Pronunciation Learning (CAPL) systems.

The level of diagnostic information provided by the MDD model depends on the level of evaluation the trained model can perform which in turn is restricted by the level of annotation of the training data. Only if the annotation is at the phoneme-level, i.e., speech is annotated with its pronounced phoneme sequence, can a phoneme-level model be trained and used to provide phoneme-level diagnostic information, e.g., the location of the mispronounced phoneme and the type of the error (substitution, deletion, or insertion).

Phoneme-level assessment is the most common assessment level used in MDD (Leung et al., 2019; Feng et al., 2020; Shahin et al., 2014; Yan et al., 2020; Harrison et al., 2009; Witt and Young, 2000; Shahin and Ahmed, 2019; Smith et al., 2017). Different approaches have been proposed to achieve phoneme-level MDD, including the scoring approach (Witt and Young, 2000; Soong et al., 2004; Zheng et al., 2007; Zhang et al., 2008; Yan and Gong, 2011; Sim, 2009; URL https://www.clapa.com/treatment/early-years-1-4/speech/; Hu et al., 2015), the rule-based approach (Abdou et al., 2006; Elaraby et al., 2016; Harrison et al., 2009; Shahin et al., 2014), the classification approach (Truong et al., 2004; Strik et al., 2007; Franco et al., 2014; Doremalen et al., 2009; Zhao et al., 2012; Wei et al., 2009; Hu et al., 2014; Baevski et al., 2020; Xu et al., 2021), and the free-phoneme recognition approach (Ye et al., 2022; Li et al., 2016b; Fu et al., 2021; Li et al., 2015).

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One of the limitations of phoneme-level MDD is that it can only diagnose categorical pronunciation errors or mispronounced phonemes that exist in the acoustic model. Uncategorical errors, where the pronounced phoneme is a distorted version of the modelled phoneme or a new phoneme borrowed from other languages, are difficult to detect with phoneme-level MDD. To handle uncategorical errors, the acoustic model needs to include phonemes from multiple languages as well as all possible pronunciation variations of each phoneme, which is infeasible. Moreover, diagnostic information obtained from the phoneme-level MDD cannot be used to construct formative feedback with corrective instruction to the learner.

In this work, we propose a low-level MDD system that detects and diagnoses pronunciation errors at the phonological level. Phonological features, such as manners and places of articulations, provide a low-level description of sound production in terms of which articulators are involved and how these articulators move to produce a specific sound. Any alteration in these features causes a pronunciation error. Therefore, accurate modelling of these features instead of phonemes can pave the way to a fully automated and interactive pronunciation assessment application where the learner receives informative and diagnostic automatic feedback not only about the existence of incorrect pronunciation, but also how the error is made. Furthermore, phonological features can be modelled solely using typically pronounced datasets which are abundantly available, unlike atypical datasets such as disordered or non-native speech. Additionally, phonological features are common across most spoken languages enabling modelling with speech corpora from multiple languages (Siniscalchi et al., 2013).

Our proposed phonological features detection model is a wav2vec2-based Sequence-to-Sequence (Seq2Seq) classification model trained using a multi-label variant of the Connectionist Temporal Classification (CTC) criteria to handle the non-mutually exclusive nature of phonological features, i.e, the same phoneme can be characterized by multiple phonological features. To validate our model, it was applied to an English L2 speech corpora collected from speakers with different native languages (L1). Additionally, we compared the detection and diagnostic accuracy of a state-of-the-art phoneme-level MDD method with our proposed phonological-level approach. We also conducted experiments to demonstrate the low-level diagnosis capability of the model and the potential formative feedback it can provide.

This paper presents three key contributions. Firstly, we established a new benchmark for phonological feature detection by utilizing the wav2vec2 speech representation upstream model. Secondly, we introduced a novel multi-label CTC approach to simultaneously learn 35 non-mutually exclusive phonological features. Lastly, we proposed a low-level MDD method based on our phonological feature detection model.

The paper is structured as follows: In Section 2, the limitations of current phoneme-level MDD approaches are discussed. Section 3 presents our proposed method, followed by a description of the speech corpora used in Section 4. The experimental setup is outlined in Section 5, while the results are demonstrated and discussed in Section 6. Finally, Section 7 concludes the paper.

2. Related work

2.1. Mispronunciation Detection and Diagnosis (MDD)

The Goodness Of Pronunciation (GOP) algorithm (Witt and Young, 2000) is the earliest and most successful phoneme-level speech assessment method utilized in several applications to measure phoneme-level pronunciation quality (Saz et al., 2009; He et al., 2015; Pellegrini et al., 2014; Mak et al., 2003; Hindi et al., 2014; de Wet et al., 2009; Luo et al., 2009). The GOP approximates the posterior probability of each phoneme by taking the ratio between the forced alignment likelihood and the maximum likelihood of the free-phone loop decoding using a Gaussian Mixture Model-Hidden-Markov Model (GMM-HMM) acoustic model. This score is then used as a confidence score representing how close the pronunciation is to the target phoneme. Subsequent methods have leveraged DNN acoustic modelling and proposed a DNN-based method to estimate the GOP (URL https://www.clapa.com/treatment/early-years-1-4/speech/; Hu et al., 2015; Sudhakara et al., 2019.

Despite its success, the GOP algorithm is very sensitive to the quality of the acoustic model used. The acoustic model affects not only the estimate of the posterior probability but also the accuracy of time boundaries obtained from forced alignment. In addition, as the decision threshold is determined using a mispronunciation dataset, it can be error specific and thus very hard to generalize to different types of pronunciation errors.

Phoneme-level error detection has also been treated as a binary classification problem, with each phoneme classified as "correct" or "mispronounced" using conventional classification methods such as SVM (Franco et al., 2014), decision tree (Truong et al., 2004), Linear Discriminant Analysis (LDA) (Strik et al., 2007), DNN (Hu et al., 2014), etc. Mel Frequency Cepstral Coefficients (MFCCs) are the most common features used with the classification methods (Truong et al., 2004). Formant frequencies and GOP scores were also utilized to classify between correct and incorrect phoneme pronunciation (Doremalen et al., 2009).

Although all these classifiers led to a significant improvement compared with confidence score methods such as GOP, they still need large amounts of accurately annotated non-native data to model the mispronounced phonemes. Moreover, the mispronounced data needs to include all possible pronunciation errors, which is usually not feasible to collect.

In Shahin and Ahmed (2019), an anomaly detection-based model was trained solely on the standard English pronunciation, namely native English speakers, and tested on foreign-accented speech and disordered speech. The method treated the mispronunciation as a deviation (anomaly) from the standard pronunciation. To detect the anomalies a One-Class SVM (OCSVM) model was trained for each phoneme using phonological features, namely manners and places of articulation. The method was shown to outperform the DNN-based GOP method in both disordered and foreign-accented speech. Recently, Xu et al. (2021) investigated fine-tuning a self-supervised speech representation pre-trained model, namely wav2vec2 (Baevski et al., 2020) to perform phoneme level binary classification of L2 speech pronunciation as correct/mispronounced. Both the scoring and classification approaches can only provide either a soft (score) or hard (binary) decision on the phoneme pronunciation without any detailed description of the type of error.

The Extended Recognition Network (ERN) method was proposed to provide more detailed descriptions of the pronunciation error. It can identify the location of the mispronounced phoneme, and the type of the error, and recognize the erroneous phoneme. Unlike forced alignment used in the scoring method which contains only the canonical phonetic transcription of the prompt word, the ERN extends the single path alignment to a network that contains the correct (canonical) path and the expected mispronounced paths on phoneme level based on phonological rules designed for a specific learning domain (Abdou et al., 2006; Shahin et al., 2014; Harrison et al., 2009).

The design of the search network is crucial for the reliability of the ERN-based MDD systems. Hand-crafted phonological rules are the most common designing criterion for the ERN (Harrison et al., 2009; Shahin et al., 2014; Smith et al., 2017; Abdou et al., 2006; Elaraby et al., 2016). Data-driven phonological approaches have also been proposed to automatically create the ERN. Lo et al. (2010) first performed an automatic phonetic alignment step between the canonical phonetic transcription and the corresponding L2 transcription then all pairs of mismatched phonemes along with their contextual phones were grouped to form the initial set of rules. Finally, a rule selection criterion was performed to select the most significant rules. In Qian et al. (2010a), the authors proposed a grapheme-to-phoneme-like model trained on phonetic transcriptions of L2 learners.

The decoding of the ERN is performed by an acoustic model that is trained either on standard pronunciation speech corpora, such as L1 native speakers (Harrison et al., 2009; Lo et al., 2010), or non-standard pronunciation speech corpora, such as L2 non-native speakers (Qian et al., 2010b; Kyriakopoulos et al., 2020), or child disordered speech (Shahin et al., 2014). As the ERN method was proposed around 20 years ago, the most common acoustic model used was the GMM-HMM acoustic model (Abdou et al., 2006; Harrison et al., 2009; Lo et al., 2010; Wang and Lee, 2012; Qian et al., 2010b). Later the DNN-HMM acoustic model was utilized (Elaraby et al., 2016; Shahin et al., 2014; Smith et al., 2017; Kyriakopoulos et al., 2020) and shown to outperform the GMM-HMM in the ERN methods specifically when trained on a small non-standard dataset (Shahin et al., 2014). Moreover, the standard acoustic model has been adapted to non-standard domains using techniques such as Maximum Likelihood Linear Regression (MLLR) for GMM-HMM-based models (Abdou et al., 2006; Wang and Lee, 2012) or transfer learning for DNN-HMM-based acoustic models (Smith et al., 2017). Qian et al. (2010b) proposed a discriminative training criterion to directly optimize the acoustic model for MDD. The authors incorporated False Acceptance Rate (FAR), False Rejection Rate (FRR), and Diagnostic Error Rate (DER) in the acoustic model objective function and trained it using L2 non-native speech corpora.

With significant improvements in End-to-End (End2End) deep learning-based acoustic models, most of the current SOTA MDD approaches have adopted free-phone recognition criteria. Unlike the ERN, free-phone recognition has no restricted search space and therefore any pronounced phoneme sequence can be captured. That is, free-phone recognition MDD systems work by estimating the pronounced phoneme sequence and the evaluation of the system is achieved by aligning the recognized phoneme sequence with the annotated one. The free-phoneme recognition model is adapted to the pronunciation learning domain by incorporating linguistic information from the prompts used in pronunciation learning which, in most cases, are pre-designed and available. This linguistic information is extracted on the character level (Feng et al., 2020), phonemic level (Ye et al., 2022; Li et al., 2016b; Fu et al., 2021; Li et al., 2015), graphemic level (Li et al., 2016b), or a combination of different levels (Li et al., 2016b).

In Leung et al. (2019) the CNN-RNN-CTC system was proposed as an early attempt of an End2End MDD method. The system has a straightforward phoneme recognition architecture trained using a combination of native and L2 speech corpora and makes use of the Connectionist Temporal Classification (CTC) loss to avoid direct alignment between the phoneme sequence and the speech signal.

Recent methods adopted an encoder–decoder mechanism to achieve MDD by using multiple encoders to encode the audio signal (Jiang et al., 2021) along with the prompt sentence phonemes (Fu et al., 2021; Ye et al., 2022), characters (Feng et al., 2020), or words (Lo et al., 2020). The encoder architecture is commonly a stacked CNN-RNN (Feng et al., 2020; Ye et al., 2022) while the decoder is mostly an attention-based network.

All the previous methods use hand-crafted features extracted from a short-time speech window, however, several recent SOTA speech recognition systems leverage the raw speech signal and use a learnable feature extraction network that can be integrated and trained within the whole network (Yan and Chen, 2021; Ravanelli and Bengio, 2018; Peng et al., 2021).

One of the shortcomings of the free-phoneme recognition method is that the phoneme sequence output is limited to the modelled phoneme set. Therefore, it is not able to detect distortion errors where the learner pronounces a distorted version of a phoneme that cannot be explicitly replaced with another phoneme within the target language phoneme set. This commonly occurs when the L2 learner is influenced by their L1 language.

In pronunciation learning, pronunciation errors made by the learner can be unpredictable. For instance, in L2 learning pronunciation errors are influenced by the proficiency of the learner and the degree of discrepancy between L1 and L2. Therefore, there are high variations in the way that the learner will adapt their pronunciation to try and match the target pronunciation. Given the limited amount of annotated mispronounced speech data available, it is infeasible to model all these variations. Existing MDD methods can only diagnose categorical pronunciations that can be modelled by the acoustic model, and struggle otherwise. Scoring approaches such as the GOP can help in detecting mispronunciations if the deviation from the correct pronunciation is high enough, however, the actual pronounced error and its type cannot be determined.

Here we have thus proposed a low-level MDD based on the phonological features. Phonological features can break down the phoneme into elementary components that form the phoneme. These features include manners and places of articulations that describe the articulators' positions and movements during sound production. There are several advantages of using phonological features in MDD: (1) the phonological features are only limited by the possible positions of the articulators and thus shared amongst most spoken languages, (2) their models can be solely trained using correctly pronounced speech which alleviates the need for annotated mispronounced data, and (3) they can provide lower-level diagnostic information describing how the error is made and which phonological feature is missing allowing more formative feedback to be provided.

2.2. Phonological feature modelling

Phonological features have been successfully utilized in various domains such as improving ASR performance (Chen et al., 2014; Lee and Siniscalchi, 2013), language identification (Siniscalchi et al., 2013), speaker verification (Qi et al., 2020) and Mispronunciation Detection and Diagnosis (MDD) (Li et al., 2016a). However, the phonological feature detection models are commonly trained at the frame level and hence frame-level labelling of the speech signal is required for all training samples. The phonological features are obtained by first performing forced alignment on a phoneme level and mapping phonemes to their corresponding features. The performance of the resultant phonological feature model is thus highly dependent on the quality of the acoustic model used in the forced alignment process. To overcome this constraint, some studies have explored using the CTC learning criteria, which eliminates the need for prior alignment between the input signal and output label (Pradeep and Rao, 2019; Karaulov and Tkanov, 2019; Qu et al., 2018; Cernak and Tong, 2018).

The phonological features are non-mutually exclusive; each phoneme can appear in multiple features. For example, the phoneme /z/ is fricative, voiced and alveolar. Consequently, modelling all features with a single model becomes a multi-label classification problem. Existing approaches address this by constructing multiple models, with each model dedicated to an individual feature (Lee et al., 2007; King and Taylor, 2000), or a set of mutually exclusive features (Abraham et al., 2016; Merkx and Scharenborg, 2018). Unfortunately, using multiple models is impractical specifically in real-time applications. In the inference time, multiple models need to be decoded which increases the application latency and consumes large memory making using phonological features unfeasible for applications that need instant feedback to be provided to the user.

In this work, we proposed a multi-label CTC method to train a single model that can detect multiple non-mutually exclusive phonological features. We also investigate using wav2vec2 speech representations to perform phonological feature detection as a downstream task.

3. Method

Two MDD systems were implemented, a conventional phoneme-based MDD similar to the wav2vec2-based model introduced in Baevski et al. (2020) and the proposed phonological(attribute)-based, with the former used as a baseline to compare the performance of our proposed

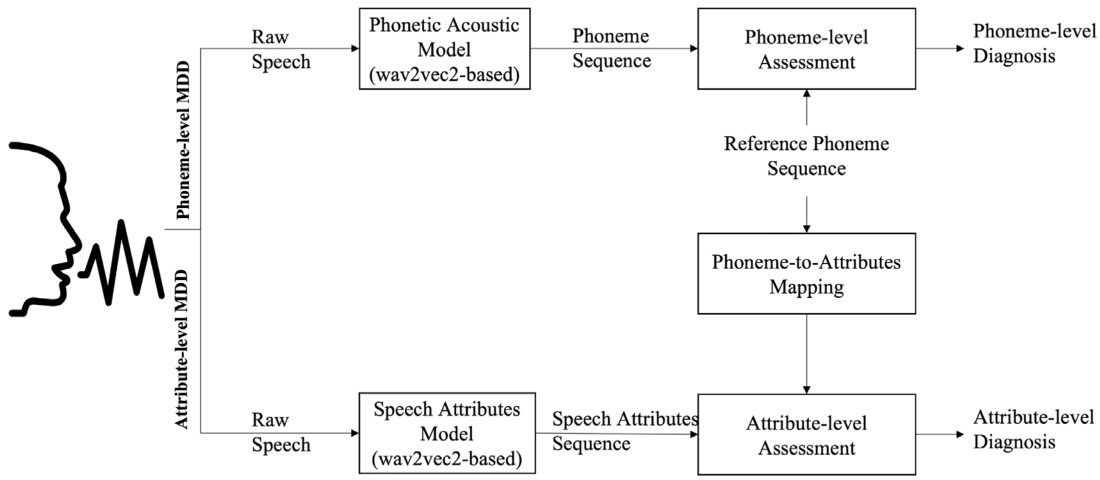

Fig. 1. A block diagram of the phoneme and phonological level Mispronunciation Detection and Diagnosis (MDD) approaches implemented. The wav2vec2-based phonetic acoustic model processes the raw speech signal and outputs a sequence of recognized phonemes. The wav2vec-based phonological features model processes the raw speech signal and produces multiple phonological feature sequences, one for each targeted feature. The assessment was performed at the phoneme level by aligning the recognized phoneme sequence with the reference one, and at the phonological features level by aligning each recognized feature sequence of $+att/-att$ with the corresponding reference phonological feature sequence.

MDD. A block diagram describing the two methods is depicted in Fig. 1. In the phoneme-based method, the raw speech signal was processed by the phonetic acoustic model to generate a sequence of recognized phonemes. The resultant phoneme sequence was then aligned with the reference (canonical) phoneme sequence to identify and diagnose the pronounced errors at the phoneme level. In the second attribute-based method, the raw speech signal was processed by the phonological features (speech attributes) model to generate multiple sequences of $+att/-att$ representing the existence or absence of each attribute. The reference attribute sequence was obtained by first mapping phoneme sequence to multiple reference phonological feature sequences of $+att/-att$. For each feature, the reference and recognized sequences were then aligned to identify the missing, inserted, or replaced features at each position.

As shown in Fig. 2, the wav2vec2 architecture was used as the backbone of both the phonological feature and the phonetic acoustic models while the Connectionist Temporal Classification (CTC) criterion was adopted to compute the training loss.

Fig. 3 demonstrates an example of MDD of three substitution errors detected in the production of the sentence “There was a change” by an L2 speaker. Here the consonant sound /r/ was pronounced as /ah/ vowel while the voiced /z/ and /jh/ sounds were replaced with the voiceless /s/ and /ch/. As seen, the phonological-feature detection model provides a detailed description of the pronunciation errors in terms of the manner and places of articulation.

3.1. Connectionist Temporal Classification (CTC)

The Connectionist Temporal Classification (CTC) loss function, initially proposed for speech recognition (Graves et al., 2006), enables learning of a Seq2Seq model without explicit alignment between the input and target sequences. Therefore, it has become the state-of-the-art learning criterion for several Seq2Seq models in various domains, including computer vision (Shi et al., 2016), handwritten text recognition (Coquenet et al., 2022), and speech recognition (Chan et al., 2016). The goal of the CTC algorithm is to compute $p(l|x)$ where $x = (x^1, x^2, x^3, \dots, x^T)$ is an input sequence of length T and $l = (l^1, l^2, l^3, \dots, l^U)$ of length U its corresponding target sequence, with $U \leq T$.

L defines a finite alphabet containing all possible labels where $l^t \in L$. CTC also defines a *blank* label which is used when no output is assigned to a specific time slot and to differentiate time slots that belong to the same label from time slots that belong to a repeated label. Hence, let $L' = L \cup blank$. The output tensor y of the network is therefore of size $T \times |L'|$ where $|L'|$ is the total number of possible labels in addition to the *blank* output. The probability of each element $i \in L'$ at time t is denoted as y_i^t, with the softmax operation performed over y^t to give $\sum_{i=1}^{|L'|} y_i^t = 1$.

If π is a label sequence of length equal to T and labels $\pi^t \in L'$, the probability of producing an output sequence π is computed as in (1)

$$p(\pi|x) = \prod_{t=1}^{T} y_{\pi^t}^t \tag{1}$$

CTC also defines a many-to-one mapping β that maps multiple label sequences of length T and labels $\pi^t \in L'$ to one label sequence l of length $U \leq T$ and labels $l^t \in L$. The mapping is performed by removing *blank* and repeated labels. For example, $\beta(a-ab-) = \beta(aa-ab) = \beta(-a-ab) = aab$ where $-$ denotes *blank*. Therefore, the probability of a label sequence l given an input sequence x can be computed as the sum of all paths mapped to l.

$$p(l|x) = \sum_{\pi \in \beta^{-1}(l)} p(\pi|x) \tag{2}$$

3.2. Connectionist temporal classification for multi-label sequences

As the phonological features are non-mutually exclusive, where each phoneme is described by more than one feature, the phonological feature detection becomes a multi-label classification problem. Each speech utterance can thus be mapped to different phonological feature sequences. The traditional CTC method can only handle single-label inputs and therefore a multi-label variant of the CTC is needed to jointly model all phonological features.

Two approaches were introduced to handle multi-label classification problem using CTC criteria, the Separable CTC (SCTC) and the Multi-label CTC (MCTC) (Wigington et al., 2019). The SCTC works by computing the CTC loss over each labelling category separately and then adding them together (i.e. multiplying the conditional probabilities) to get the target loss. In Li and Hasegawa-Johnson (2020), the authors used the SCTC criteria to train the CTC-based model to recognize both phones and tones in multilingual and cross-lingual scenarios.

On the other hand, the MCTC first computes the probability of each target element from its categorical components and then computes the CTC loss over the target sequence (Wigington et al., 2019). The MCTC

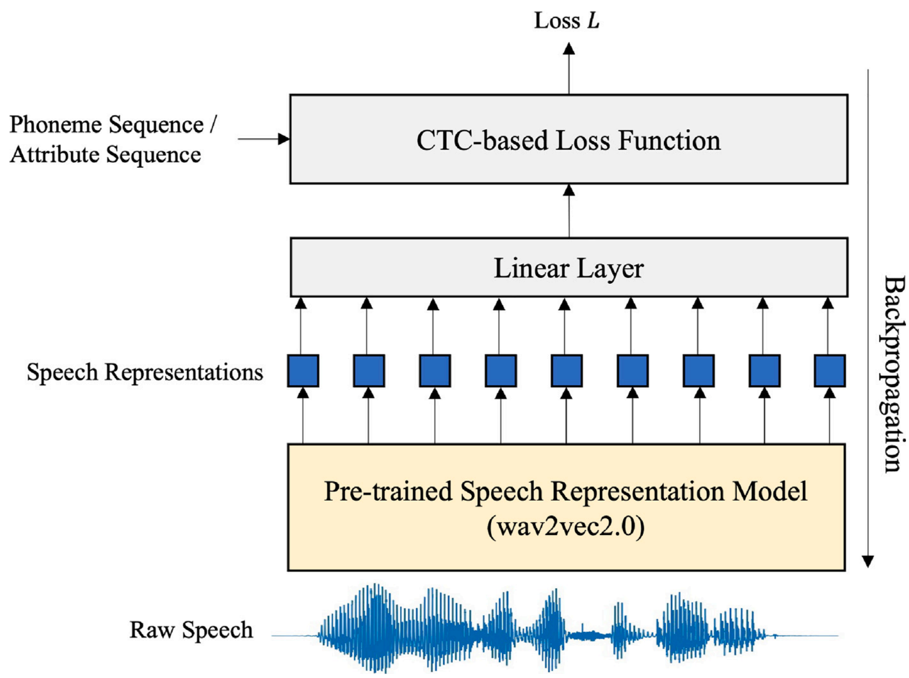

Fig. 2. The architecture of the phoneme/phonological model. The speech representations generated by the pre-trained wav2vec2 model were passed to a linear layer with number of nodes equals to the number of target phonemes/phonological-features. The model was fed by the raw speech signal and the CTC-based loss function was calculated by considering all possible alignments between the input signal and the target phoneme/phonological-features sequence.

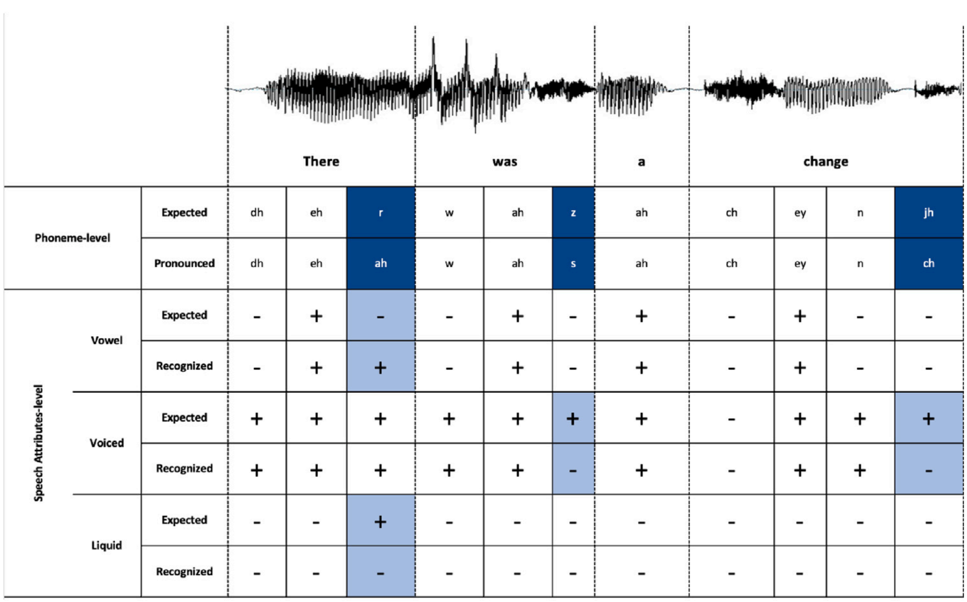

Fig. 3. Mispronunciation detection and diagnosis example of sentence *"There was a change"*. /r/, /z/, and /jh/ phonemes are pronounced as /ah/, /s/, and /ch/ respectively. /r/ is $-vowel$ and $+liquid$ while /ah/ is $+vowel$ and $-liquid$. Both /z/ and /jh/ are $+voiced$ while their associated erroneous phonemes /s/ and /ch/ are $-voiced$. The proposed MDD breaks down the pronunciation error into elementary components (phonological features) and identifies the incorrect feature.

approach has been successfully applied to polyphonic music audio to detect pitch classes (Weiss and Peeters, 2021) and to handwritten text recognition (Wigington et al., 2019).

Wigington et al. (2019) discuss the two approaches for the multi-label CTC. One major issue with SCTC is that each category is treated separately and therefore it is not guaranteed that at each frame the correct combination of components exists.

In this work, we adopted the SCTC approach as the objective is the classification of the separate phonological feature categories. However, to maintain the alignment between components, one blank node was shared among all categories. The reasoning behind using a shared blank token is that in phonological feature modelling, as the categories are binary representations of each feature, each phoneme has one and only one representation in each category. The blank token over all categories thus represents the silence frames or the transition from one phoneme to another. Using a shared blank, increases the likelihood that all categories will produce a blank at the same time frame. We refer to this approach as Separable CTC with Shared Blank (SCTC-SB).

In the multi-label scenario, each input sequence has multiple target sequences with labels derived from different alphabet categories. N categories $C = \{C_1, C_2, C_3, \ldots, C_N\}$ are then defined, where C_i represents the alphabet of category i. For any input x with target sequence l, each

Table 1
The 35 learned phonological features including manners and places of articulation and other phonological features such as voiceness (Chomsky and Halle, 1968).

Manners	Places	Others
Consonant, sonorant, fricative, nasal, stop, approximant, affricate, liquid, vowel, semivowel, continuant	Alveolar, palatal, dental, glottal, labial, velar, mid, high, low, front, back, central, anterior, posterior, retroflex, bilabial, coronal, dorsal	Long, short, monophthong, diphthong, round, voiced

element in l is then decomposed into N components representing the N categories where:

$$l^t = (l_1^t, l_2^t, l_3^t, \ldots, l_N^t) \tag{3}$$

Therefore, the input x of length T and target sequence l of length $U \leq T$ can be represented in N sequences of l_i all of length U.

The network output tensor y will be of size $T \times (\sum_{i=1}^{N} |C_i| + 1)$ where $|C_i|$ is the number of elements in category C_i plus the shared blank node. Let $C_i' = C_i \cup blank$, then the probability of output element j of category C_i' at time t is denoted as $y_{i,j}^t$. In this case, the softmax function is applied over the components of C_i', hence $\sum_{j=1}^{|C_i'|} y_{i,j}^t = 1$.

The probability of each category is then computed separately as:

$$p(l_i|x) = \sum_{\pi_i \in \beta_i^{-1}(l_i)} p(\pi_i|x) \tag{4}$$

where π_i is a label sequence of length T with $\pi_i^t \in C_i'$ while β_i is a many-to-one mapping defined for each category i that maps π_i to l_i, and the probability of the output path π_i is computed as:

$$p(\pi_i|x) = \prod_{t=1}^{T} y_{i,\pi_i^t}^t \tag{5}$$

The final objective function is then computed as the multiplication of the label probabilities of all categories:

$$p(l|x) = \prod_{i=1}^{N} p(l_i|x) \tag{6}$$

3.3. Phonological feature modelling

$N = 35$ phonological features (speech attributes) were adopted representing the manners and places of articulation along with other phonological features as listed in Table 1. These phonological features were selected so that each phoneme has a unique binary representation in terms of the 35 features. The 35 phonological features were jointly learnt using the proposed SCTC-SB criterion. A category for each feature (*att*) was defined with items $C_i = +att, -att$. The category that represents the nasal feature, for example, has possible outputs of $+nasal, -nasal$. Therefore, the number of network outputs is equal to 71, where 35 nodes represent the existence of each feature ($+att$), 35 nodes represent the absence of each feature ($-att$), and one node represents the blank output that is shared among all categories.

In the training phase, the phoneme sequence l of each utterance was mapped to 35 target sequences l_i of $+att, -att$ symbols, one for each feature as shown in Fig. 4 for an example phrase *"How old are you?"*. The CTC loss of each category was then computed using (5) and the final loss function for the utterance was calculated according to (6) by multiplying the 35 CTC losses of all categories producing the phoneme sequence loss.

In the inference phase, for any speech input x of length T, the most probable labelling of each feature i was obtained by applying an arg max function over the output nodes representing items in each category C_i' as follows

$$h_i(x) = \text{argmax}_j \, y_{i,j}^t, \quad j = 1, 2, \ldots, |C_i'| \tag{7}$$

Table 2
Distribution of the utilized datasets.

	Name	Type	Hours	Speakers
Training	TIMIT	Native	3.9	462
	LS-clean-100	Native	100	251
	TIMIT+L2	Native+Non-Native	6.5	480
Testing	LS-clean-test	Native	4.1	40
	LS-other-test	Native	4.4	33
	TIMIT-test	Native	1.4	168
	L2-Scripted	Non-Native	0.11	6
	L2-Suitcase	Non-Native	0.87	6

The output is a sequence of $+att, -att$ and blank tokens of length T. Finally, the repeated tokens were merged, and all blank tokens were removed to get the final output sequence of length $U \leq T$ using predefined category mapping β_i.

4. Speech corpora

Three English speech corpora were employed in this paper. The native Librispeech (LS) (Panayotov et al., 2015) corpus, which consists of audiobooks from the Librivox project (Kearns, 2014), the native TIMIT dataset (Garofolo et al., 1993), and the non-native second language learning L2-ARCTIC corpus (Zhao et al., 2018). The LS corpus is a frequently utilized open-access database for evaluating diverse speech-processing tasks. On the other hand, the TIMIT dataset is a well-designed, small dataset that includes phonetically balanced recordings from various dialects. Additionally, the non-native L2-ARCTIC corpus is a relatively recent speech corpus that has become the standard dataset for evaluating MDD systems.

The LS dataset has "clean" and "other" versions, referred to as LS-clean and LS-other, respectively. The "other" portion of the LS dataset, LS-other, contains challenging speech data that has lower recognition accuracy compared to the "clean" portion (Panayotov et al., 2015).

On the other hand, L2-ARCTIC was collected from 24 non-native English speakers equally distributed over 6 native languages, namely Arabic, Hindi, Korean, Mandarin, Spanish, and Vietnamese. The L2-ARCTIC contains two subsets, scripted and spontaneous, referred to as L2-Scripted and L2-Suitcase respectively. The scripted one consists of 27 h of speech, of which only 3.5 h were manually annotated at the phoneme level. In contrast, the spontaneous speech subset contains 26 min of speech where each speaker recorded around one minute. Unlike the scripted speech, the whole spontaneous speech utterances were manually annotated at the phoneme level. The same speakers recorded both scripted and spontaneous subsets. The L2-ARCTIC was further split into training and testing subsets following the same split as in Ye et al. (2022), where six speakers, one from each L1 language, formed the test set while the other 18 speakers were used for training.

The TIMIT corpus, 100 h of the LS-clean, referred to as LS-clean-100, and a combination of TIMIT and L2-ARCTIC corpora, referred to as TIMIT+L2, were used separately for training, developing and testing of the phonological feature detection and the phoneme recognition acoustic models. While the L2-ARCTIC was for the evaluation of the MDD system. Table 2 summarises the utilised datasets.

Around 15% of the L2-ARCTIC has manual annotations at the phoneme level describing each pronunciation error by indicating the pronounced phoneme, if pronounced, and if the error is an insertion (I), deletion (D), or substitution (S). Table 3 shows the number of each type of error, along with the number of correctly pronounced phonemes (C), in the training and testing subsets of the L2-Scripted and L2-Suitcase datasets.

The CMU dictionary (CMUdict) (Weide, 1998) was used to obtain the phonetic transcription of the LS orthographic transcription. To match the phoneme set of LS and L2-ARCTIC, the TIMIT phonemes were converted from 61 to 39 following the mapping proposed in Wigington et al. (2019). However, /zh/ was not mapped to /sh/ as the

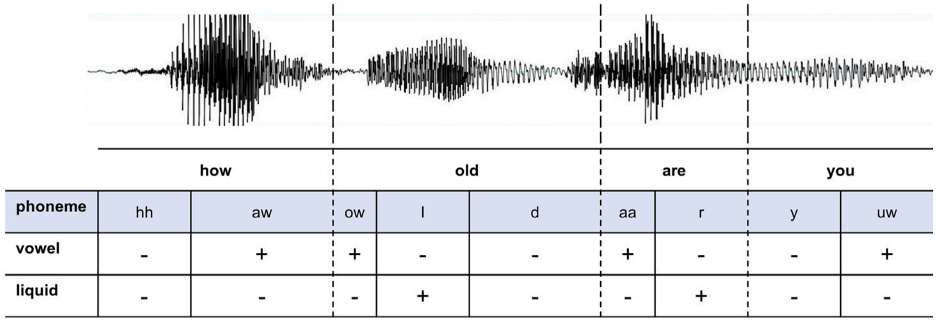

Fig. 4. Example feature mapping for the phrase 'How old are you?' showing the ground truth phoneme sequence and the corresponding vowel and liquid features labels. Here the vowel feature is present for /aw/, /ow/, /aa/, and /uw/, while the liquid is present for /l/ and /r/ consonants.

Table 3
The number of Correct (C), Substituted (S), Inserted (I), and Deleted (D) phonemes in the L2ARCTIC speech corpus.

	Subset	C	S	I	D
L2-Scripted	Training	79 864	10 474	772	2437
	Testing	28 331	3198	214	939
L2-Suitcase	Training	5736	1032	55	290
	Testing	2333	438	28	137

former is $+voiced$ while the latter is $-voiced$ and merging them can confuse the *voiced* feature model. All silence labels were further removed leaving the silence frames to be handled by the blank label.

5. Experimental settings

5.1. Experiments

First, we performed a number of experiments to assess the performance of our proposed phonological feature detection method followed by a comparison between the phonological-level MDD and the phoneme-based MDD. We conducted the following experiments:

1. To investigate the influence of the pre-trained model size and the domain of the speech corpora used in the pre-training process, we compared the performance of the three wav2vec2-based pre-trained models with respect to phonological feature recognition: wav2vec2-base, wav2vec2-large, and wav2vec2-large-robust.
2. To explore the robustness of the proposed phonological features recognition approach, we compared its performance when applied to different domains (LS and TIMIT) as well as when used with out-of-domain data (native/non-native).
3. To demonstrate the effectiveness of the proposed wav2vec2-based phonological feature detection model, we compared its performance to an existing baseline based on the Deep Speech 2.0 (Amodei et al., 2016) model.
4. We finally assessed the effectiveness of our phonological feature recognition approach when used for Mispronunciation Detection and Diagnosis (MDD) by applying the method to L2 speech and comparing it to phoneme-level MDD.

5.2. Training procedures

As aforementioned, the core model of the architecture is the self-supervised pre-trained wav2vec2 model. Fig. 5 depicts the block diagram of the training procedure of our proposed model. The wav2vec2 model consists of a multi-layer CNN encoder that generates the latent representations followed by a transformer module with multiple blocks and multiple attention heads. A linear layer was added on top of the transformer module with the number of nodes representing the number of classes. The number of classes used to train the SCTC-SB-based phonological feature model was 71: 35 for the existence and 35 for the absence of each feature plus one node for the *blank* output. For the CTC-based phoneme recognition model, 40 output nodes were used representing the 39 phonemes in addition to a *blank* node.

Except for the CNN encoder layer, the whole network was then fine-tuned to minimize either the SCTC-SB or CTC loss using back-propagation. As the feature extraction layer was already well-trained during pre-training, its parameters were fixed during the fine-tuning process. Furthermore, SpecAugment (Park et al., 2019) was applied to the output of the CNN encoder to add more variations to the training data. The performance of three pre-trained models, namely wav2vec-base, wav2vec-large (Baevski et al., 2020), and wav2vec-large-robust (Hsu et al., 2021) were compared. The first two models were pre-trained on 960 h of read-out books dataset, Librispeech, while the latter one was pre-trained on multiple datasets from different domains, including read-out books and telephone conversations. The wav2vec-base model has 95M parameters while both wav2vec-large and wav2vec-large-robust have 317M parameters.

AdamW optimization (Loshchilov and Hutter, 2017) was utilized for all experiments with a 0.005 weight decay and 0.0001 initial learning rate. The batch size was fixed to 32, and the fine-tuning ran for 30 epochs. 10% of the iteration steps were consumed in the warmup phase to reach the initial learning rate.

5.3. Evaluation metrics

In this work, we utilized several performance metrics to evaluate and compare the developed models based on their different tasks

5.3.1. Phonological feature recognition performance

For the phonological feature recognition model, as the output is a binary sequence of $+att/-att$ symbols, we used the traditional error rate derived from the Levenshtein distance metric (Levenshtein et al., 1966). The Levenshtein distance metric works by measuring the difference between two sequences in terms of the number of Insertion (I), Deletion (D), and Substitution (S) edits. Therefore, the Feature Error Rate (FER) was computed as follows:

$$FER = \frac{S + D + I}{N} \tag{8}$$

where I, D, and S were estimated by comparing the recognized sequence to the reference sequence and N is the total number of reference symbols. In some experiments, we used Accuracy (ACC) instead of error rate which is simply computed as $1 - ErrorRate$.

Furthermore, for the phonological feature recognition task, we calculated the Precision (PRE), Recall (REC), and $F1$ score of each feature as follows:

$$\begin{aligned} REC &= TP/(TP + FN) \\ PRE &= TP/(TP + FP) \\ F1 &= 2 \times \frac{PRE \times REC}{PRE + REC} \end{aligned} \tag{9}$$

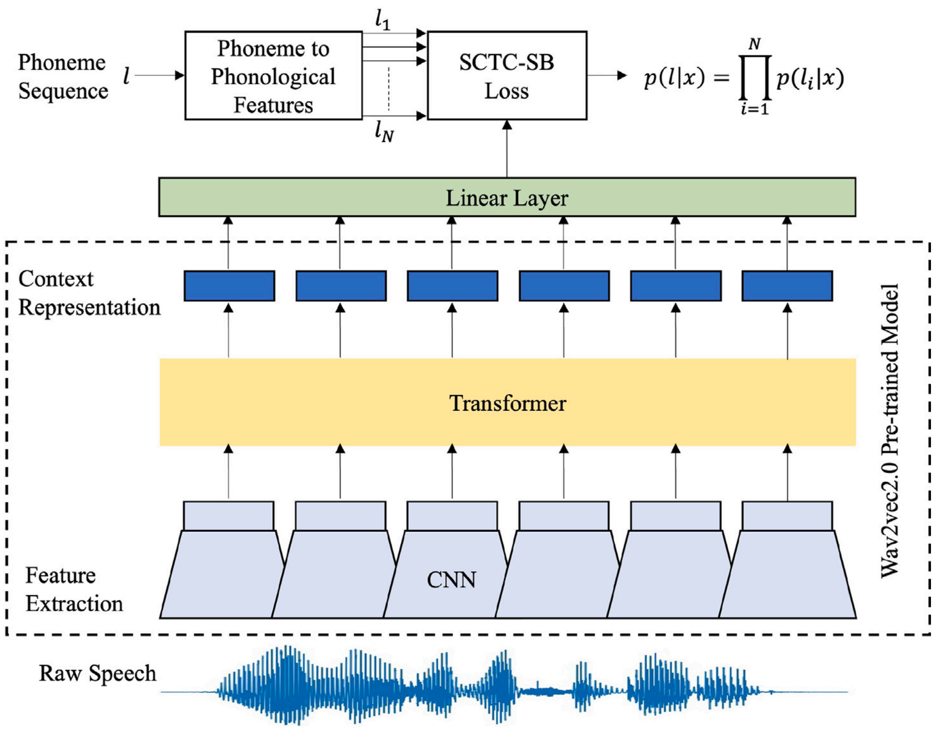

Fig. 5. The training procedure for the proposed phonological feature recognition model. A linear layer was added on top of a wav2vec2-based pre-trained model. For each speech utterance, the linear layer converts the corresponding phoneme sequence to N binary sequences of phonological features. The CTC loss was then computed for each feature sequence. Finally, the SCTC-SB loss was computed by multiplying all phonological features' CTC losses.

where TP, FP, and FN are the True-Positive, False-Positive, and False-Negative, respectively. The positive and negative here refer to the generation of the $+att$ or $-att$ symbols.

5.3.2. Mispronunciation Detection and Diagnosis (MDD) performance

For the evaluation of the MDD task, we used metrics as proposed in Shi et al. (2016) at the phoneme-level. Firstly, we counted the occurrences of the following:

1. True-Acceptances (TA), which represent the number of times a recognized phoneme matches a correctly pronounced phoneme.
2. True-Rejections (TR), which represent the number of times a recognized phoneme is different from a mispronounced phoneme.
3. False-Acceptances (FA), which represent the number of times a recognized phoneme matches a mispronounced phoneme.
4. False-Rejections (FR), which represent the number of times a recognized phoneme is different from a correctly pronounced phoneme.

The TRs were further split to Correctly Diagnosed (CD) when the recognized phoneme matches the phoneme that was pronounced, and Diagnosis Error (DE) otherwise. We then used these counts to estimate the False Acceptance Rate (FAR), the False Rejection Rate (FRR), and the Diagnostic Error Rate (DER) as follows:

$$\begin{aligned} FAR &= FA/(FA+TR) \\ FRR &= FR/(FR+TA) \\ DER &= DE/(CD+DE) \end{aligned} \tag{10}$$

Similar metrics were used for the phonological-level MDD. For instance, if the phoneme /s/ was mispronounced as /z/, this was considered a mispronunciation of the *voiced* feature only and correct pronunciation for all other phonological features.

6. Results and discussion

6.1. Phonological feature recognition

6.1.1. Comparison of pre-trained wav2vec2 models

The goal of this experiment was to explore how the size of the model parameters and the nature of the training data impact the performance of phonological feature detection. Here, multi-label SCTC-SB phonological feature recognition models were trained by minimizing the SCTC-SB loss function in (7) for all binary groups of the 35 phonological features listed in Table 1. At inference time, a sequence of 35 $+att/-att$ tokens was produced for each speech file representing a sequence of the existence/absence of each phonological feature. The phonological feature recognition accuracy (ACC), precision (PRE), recall (REC) and F1 metrics were used for performance evaluation and model comparison.

Fig. 6 shows the $F1$ and ACC metrics of the 35 phonological features as recognized by the phonological feature recognition model when fine-tuned with the TIMIT training set and tested against the TIMIT test set. Here the performance of three wav2vec2-based pre-trained models, namely wav2vec2-base, wav2vec2-large, and wav2vec2-large-robust were compared.

Although both base and large models were pre-trained on the same dataset, the large model performed better than the base one. This demonstrates that increasing the model capacity improves the phonological features recognition accuracy. On the other hand, the wav2vec2-large-robust model consistently slightly outperformed both base and large models for nearly all phonological features. The average accuracies of the three models were 96.6 ± 0.73, 96.8 ± 0.71 and 97 ± 0.65 for the base, large and large-robust models, respectively. The large-robust model was pre-trained on data from different domains to be more robust against out-of-domain data (Hsu et al., 2021). Similar behaviour was obtained when fine-tuning the three pre-trained models for the phoneme recognition task using the CTC criterion. The PERs of the base, large, and large-robust models were 8.1, 7.8, and 7.3 respectively.

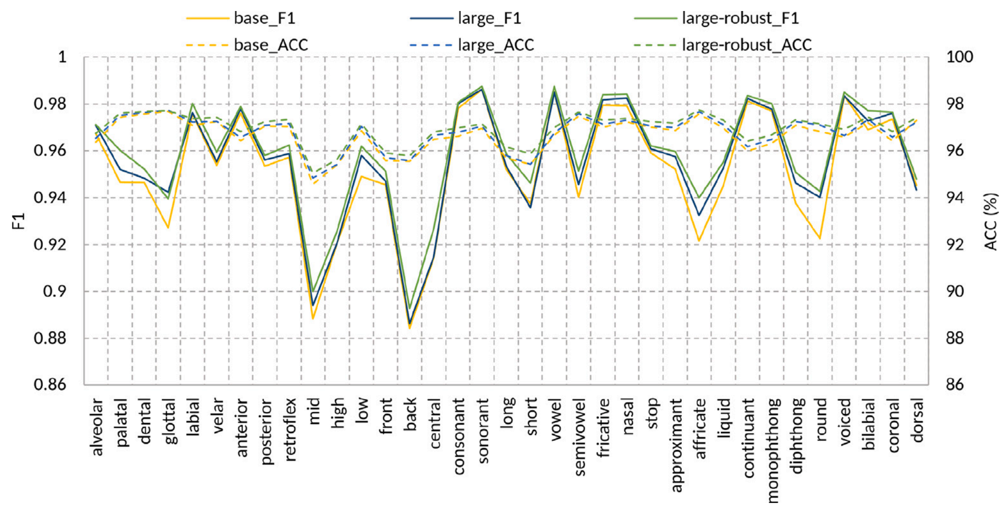

Fig. 6. The performance of three wav2vec2-based pre-trained models for the phonological feature recognition task. The plots include the F1 (solid) and ACC (dashed) metrics (%) of the 35 phonological features for the TIMIT complete test set obtained from a model fine-tuned with the TIMIT training set. The base model has the lowest performance while the large-robust performed consistently higher for all features.

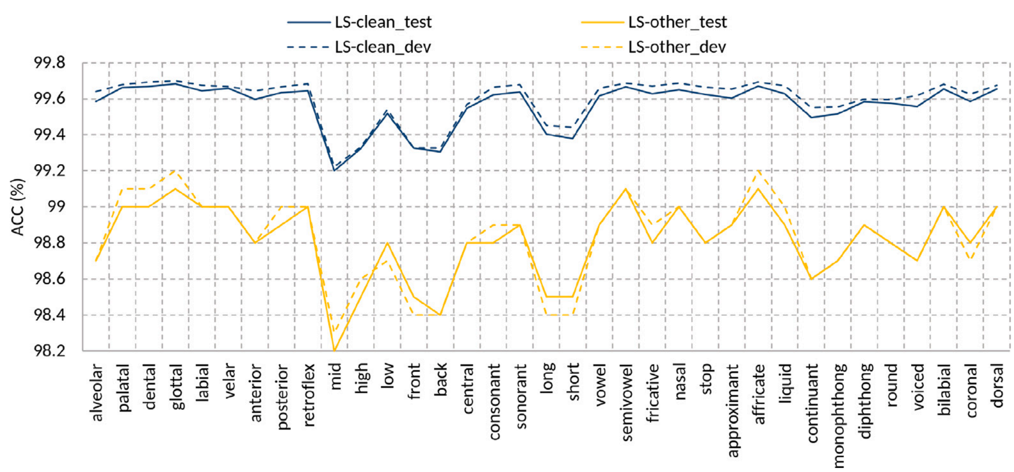

Fig. 7. The 35 phonological feature recognition accuracies (%) of the multi-label SCTC-SB model trained on the LS-clean-100 datasets and tested against the "clean" and "other" test and development sets. The "other" refers to a more challenging portion of the LS corpus. When applied to LS-other data, the model performance dropped consistently across all features with ratios ranging from 31%–45%. Model performance declined the most for features associated with vowels, including *high, front, mid*, and *back*. On the other hand, features like *fricative, stop, alveolar*, and *dental* were less impacted.

In the subsequent experiments, the pre-trained wav2vec-large-robust model was used in the phoneme and phonological feature recognition downstream tasks.

6.1.2. Performance of phonological feature model over different domains

In this experiment, we investigated the robustness of the proposed model against challenging audio, like the "other" portion of the librispeech corpus (LS-other), and in the case of domain shift from native to non-native speech. Fig. 7 shows the 35 phonological feature recognition accuracies of the proposed multi-label SCTC-SB model trained on the LS-clean-100 datasets and tested against the clean and other test and development sets, LS-clean-test, LS-clean-dev, LS-other-test, and LS-other-dev. The figure shows that the accuracies of all phonological features of the clean test and development sets are above 99%. However, applying the model to the more challenging LS-other speech datasets causes degradation across all phonological features ranging from 31% up to 45%. The highest drop in the performance occurred in the vowel-related features such as *high*, *front*, *mid* and *back* while the *fricative*, *stop*, *alveolar* and *dental* were less affected features.

To demonstrate the effect of the domain-mismatch between training and testing data, two multi-label SCTC-SB models were trained on LS-clean-100 and the TIMIT training set and tested against the TIMIT test set. As shown in Fig. 8, the in-domain experiment, where the model was trained and tested on TIMIT speech corpus, achieved the lowest Feature Error Rate (FER) of around 2.3% for dental, glottal, semivowel, and affricate features while the highest FER of 4.8% was obtained by the mid feature followed by high, back and short features. On the other hand, the model accuracy degradation ranged from 22% to 33% when trained on out-of-domain data, LS-clean-100.

Fig. 9 shows how the phonological features recognition accuracy of models fine-tuned on native (TIMIT) and native+non-native (TIMIT-L2) datasets were impacted when tested on non-native sets, L2-Scripted and L2-Suitcase. For the L2- Scripted test set, it is evident that adding non-native data to the training set significantly improves the recognition accuracy of all phonological features. However, for L2-Suitcase, not all features show the same improvement when using non-native data in the model's fine-tuning. The glottal, retroflex, nasal and diphthong features achieved almost the same accuracy with and without non-native data. Moreover, accuracies of labial and semivowel degraded slightly with non-native data.

Fig. 10 summarizes the precision, recall, and F1 quartiles of different pairs of training and testing sets. It is noticeable that the performances of both TIMIT and L2-Scripted with a multi-label SCTC-SB

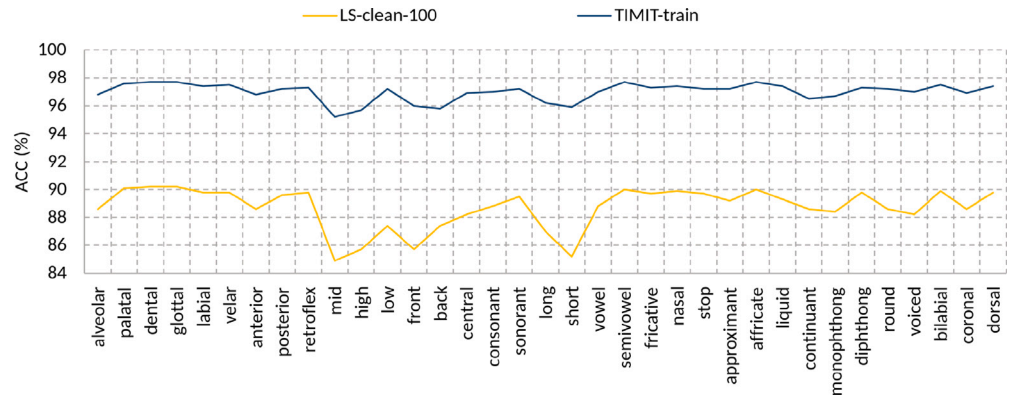

Fig. 8. The effect of domain-mismatch between training and testing data. The plots show the ACC (%) of the 35 phonological features for multi-label SCTC-SB models trained on the LS-clean-100 and TIMIT training set and tested with the TIMIT test set. Testing the model with the out-of-domain data causes a relative degradation in the ACC ranging from 22% to 33%.

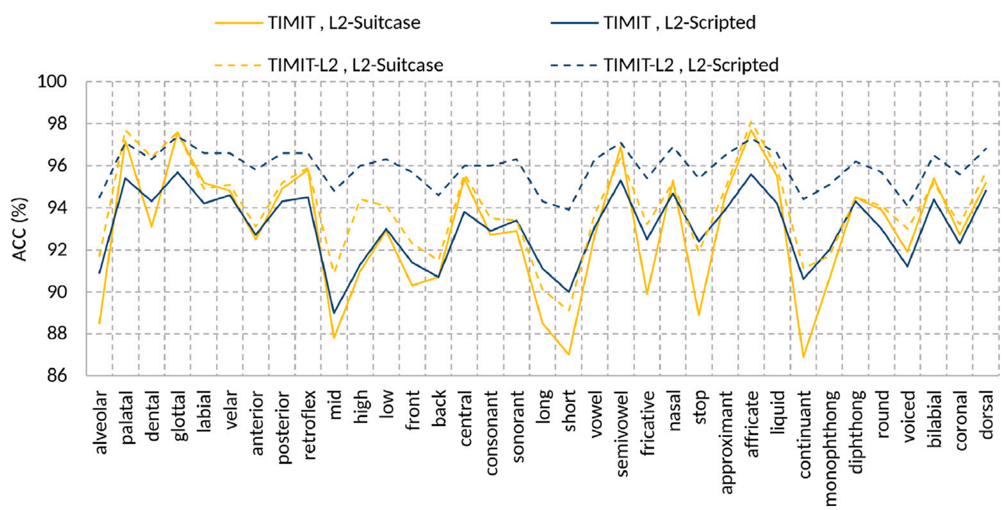

Fig. 9. The accuracy of phonological features recognition when used with non-native test sets. Here multi-label SCTC-SB models are fine-tuned with native (TIMIT) and native+non-native (TIMIT-L2) datasets and tested with non-native L2-Scripted and L2-Suitcase datasets. The L2-Scripted data achieves significant improvements when adding L2 data to the training set over all features.

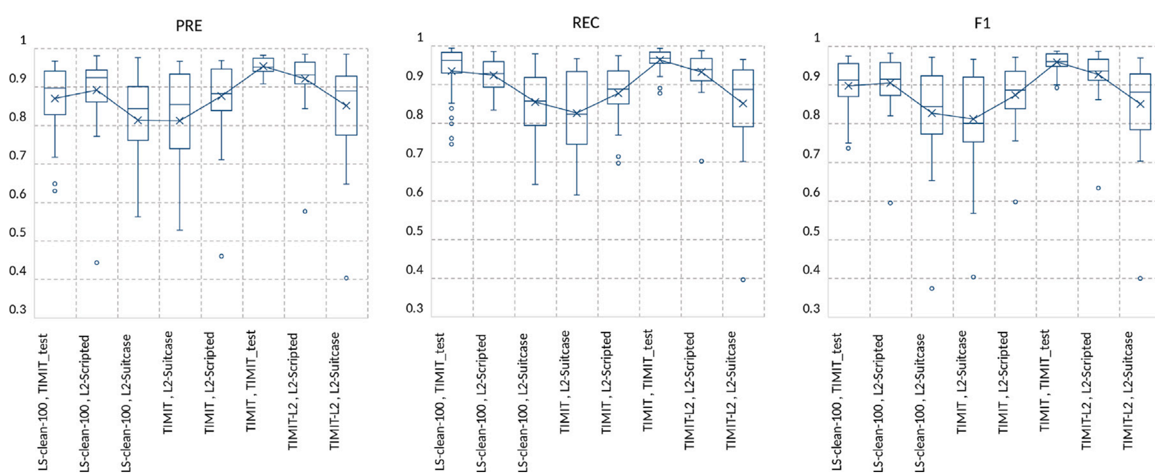

Fig. 10. A comparison of the phonological feature recognition performance metrics across different pairs of training and testing datasets. Both TIMIT and LS-clean-100 are native English data, while L2-Scripted and L2-Suitcase are non-native English scripted and spontaneous speech, respectively. The boxplots show the precision (PRE), recall (REC), and F1 quartiles of the different (training, testing) data pairs. The spontaneous L2-Suitcase test set achieves lower PRE, REC, and F1 compared with the L2-Scripted test set over all training sets.

Table 4
Evaluation results for the non-native test data using phonetic acoustic models trained on native and non-native datasets for a phoneme-level MDD task. Using a combination of native and non-native training data achieved the lowest DER over the two test sets.

Train data	Test data	FA	FR	TA	TR		FRR (%)	FAR (%)	DER (%)
					CD	DE			
LS-clean-100	L2-Scripted	2686	2261	23 920	1077	497	8.64	63.05	31.58
	L2-Suitcase	345	256	1884	191	66	11.96	57.31	25.68
TIMIT	L2-Scripted	1649	4808	21 300	1896	715	18.42	38.71	27.38
	L2-Suitcase	209	455	1649	264	129	21.63	34.72	32.82
TIMIT-L2	L2-Scripted	1683	1899	24 079	2170	407	**7.31**	**39.51**	**15.79**
	L2-Suitcase	155	268	1829	370	77	**12.78**	**25.75**	**17.23**

model trained on LS-clean-100 were comparable despite both TIMIT and LS-clean-100 being native English data. To further understand this, we re-examined the commonalities between TIMIT and L2-Scripted that are absent in LS data. We found that both TIMIT and L2-Scripted were manually transcribed at the phonemic level, whereas LS data was transcribed using a pronunciation dictionary. To determine if this factor contributed to the degradation in phonological feature detection accuracy when applying the LS-clean-100 model to TIMIT, we obtained the phonemic transcription of the TIMIT training subset using the same CMUdict dictionary used for LS data transcription. We then trained a phonological feature detection model and tested it on the manually transcribed phonetic data of the TIMIT test subset (TIMIT-test).

We found that the average phonological feature detection accuracy significantly dropped from $97\% \pm 0.65$ when the model was trained on the manual phonemic transcription of the TIMIT data to $88\% \pm 1.7$ when the CMUdict was used. This degradation closely mirrors the one observed when using a model trained on LS-clean-100, indicating that the decline in accuracy is due to the mismatch between the phonemic transcription obtained from CMUdict, which reflects the standard pronunciation of each word, and the actual pronunciation of the speakers. This discrepancy likely arises because the speakers' pronunciations may deviate from the standard due to regional accents, individual speech patterns, or other linguistic factors.

Moreover, although both L2-suitcase and L2-Scripted were recorded by the same speakers, the L2-Suitcase seems more challenging than the L2-Scripted. This is explained by the L2-Suitcase recordings' longer duration and spontaneous nature compared to the short and scripted L2-Scripted dataset.

6.1.3. Comparison with existing work

Most of the existing work on phonological feature detection reports frame-level performance. However, the proposed model is based on CTC, which outputs a sequence of symbols. As the CTC blank node allows the model to output no label in some frames when uncertain, obtaining frame-level alignment from the CTC model is not accurate. Consequently, it is hard to make a fair comparison with a frame-level phonological feature detection system. Therefore, the proposed SCTC-SB model was compared with two CTC-based systems. Both systems were based on the Deep Speech 2 model (Amodei et al., 2016) and trained using CTC. The first system was proposed in Cernak and Tong (2018) for nasality detection where the output is a sequence of +nasal,-nasal tokens. The model achieved an error rate of 4.4%. The second system, introduced in Pradeep (2018), discriminates between five manners of articulations, namely vowel, semi-vowel, nasal, stop, and fricative. The system achieved an error rate of 2.7%. Both systems were trained and tested on the LS-clean-100 and LS-clean datasets, respectively. As shown in Fig. 8, when the proposed SCTC-SB system was trained and tested on the same dataset, the error rate for all phonological features was less than 1%, significantly outperforming both baseline systems.

6.2. Mispronunciation Detection and Diagnosis (MDD)

The next experiments aim to demonstrate the effectiveness of the proposed phonological-based MDD in terms of detection and diagnosis accuracies. For comparison, we implemented a SOTA phoneme-level MDD and applied it to the same native and non-native speech corpora.

6.2.1. Phoneme-level MDD

In this experiment, MDD at the phoneme level was performed using three phonetic acoustic models trained on the LS-clean-100, TIMIT, and TIMIT-L2 datasets. Table 4 summarizes the evaluation results of the different models tested with L2-Scripted and L2-Suitcase test sets. The table shows that the model fine-tuned with the LS-clean-100 dataset has the highest FARs of 63% and 57% for L2-Scripted and L2-Suitcase respectively, and an FRR of 8% for both datasets. In contrast, the model fine-tuned with the TIMIT dataset achieved a better balance between FRR and FAR. Adding L2 data to the TIMIT training dataset significantly reduced the FRR from 18% and 21%, when using only the TIMIT training set, to 7% and 12% for the L2-Scripted and L2-Suitcase test sets, respectively. Furthermore, the model fine-tuned with the TIMIT-L2 dataset achieved the lowest DERs of 15% and 17% for L2-Scripted and L2-Suitcase test sets, respectively. These results suggest that by increasing the amount of native data in the training of the phonetic acoustic model, the model's tendency to accept pronunciation errors (FAR) is increased significantly. By comparing the TIMIT and TIMIT-L2 models, it is noticeable that although both models can detect the existence of error with a comparable FAR, adding in-domain L2 data improved the ability of the model to Correctly Diagnose (CD) the error and significantly reduce the DER.

6.2.2. Phonological-level MDD

In this experiment, the three datasets, LS-clean-100, TIMIT, and TIMIT-L2 were used to train three SCTC-SB phonological features models to detect the existence or absence of 35 phonological features listed in Table 1. The output of each model was a sequence of $+att/-att$ for each input speech signal. The evaluation was performed for each feature separately by aligning each output sequence with the corresponding reference sequence obtained by mapping the target phoneme sequence to the target phonological feature sequence. The FAR, FRR, and DER were computed as explained in Section 5.3.

Fig. 11 depicts boxplots summarizing the FAR, FRR, and DER of the 35 phonological features MDD obtained using models trained on the three mentioned training sets and tested using the L2-Scripted and L2-Suitcase test sets. The results show that the FAR of most phonological features was less than 30% when using the purely native model trained on LS-clean-100 which is significantly lower than the phoneme-level counterpart FARs of 57% and 63% for L2-Suitcase and L2-Scripted respectively (see Table 4). The DERs of the three models tested with L2-Scripted for all phonological features lies below 10% which is also significantly lower than the phoneme-level DERs of 31%, 27%, and 15% of the LS-clean-100, TIMIT, and TIMIT-L2 respectively.

Detailed results for each feature obtained from the TIMIT-L2 model are shown in Fig. 12 for both L2-Scripted and L2-Suitcase test sets. The

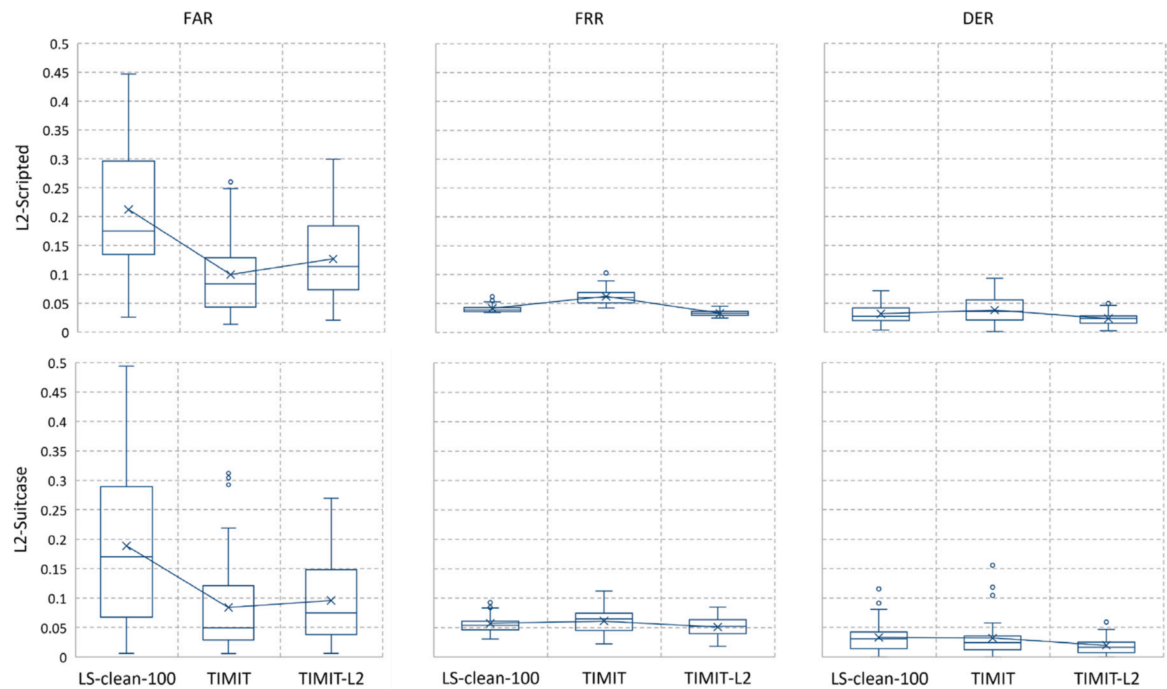

Fig. 11. Evaluation of three phonological features models in performing phonological-level MDD of two test sets, L2-Scripted and L2-Suitcase. Two models were trained on native speech corpora, namely LS-clean-100, and TIMIT, while the third model was trained on a combination of native and non-native data TIMIT-L2.

phonological features are sorted from lowest to highest FAR values. It is obvious that there are high variations in the FAR among phonological features while the DER and FRR are more consistent. The results demonstrate also that in both L2 test sets, all the phonological features have achieved FAR, FRR, and DER lower than the phoneme level equivalents (shown as straight lines).

6.2.3. Analysis of phonological-level MDD

To better understand the benefits and limitations of the phonological feature model in MDD, the phoneme-based and phonological feature-based models were used to detect 34 common substitution errors obtained from the manual annotation of the L2-Scripted dataset (e.g., /t/ as /d/ or vice versa). A complete pronunciation error matrix for L2 speakers is depicted in Fig. 13.

As shown in Tables 5 and 6, we calculated the phoneme-level FAR for substitution errors and the FRR for correct pronunciations of either phoneme for consonants and vowels common errors respectively. Similarly, we computed the FAR and FRR at the phonological level for all distinctive phonological features. Distinctive phonological features are those that differentiate one phoneme from another; for example, /p/ and /b/ share all features except *voicing* (/p/ is voiceless while /b/ is voiced). In the table, we present the phonological feature with the lowest FAR for each confused pair of phonemes. These results were derived from an in-domain scenario using a model trained on the TIMIT-L2 training set and tested on the L2 test set. We found that in 29 out of the 34 confused phonemes, at least one phonological feature provided better discrimination than the phoneme-level model, with an average FAR improvement of 16.5% ± 28. For instance, the *back* feature reduced the FAR from 80% with the phoneme model to 35% for discriminating between /ah/ and /ih/. Generally, all vowel confusions could be better discriminated by at least one phonological feature compared to the phoneme model, whereas some consonant confusions, such as /s/ and /sh/, proved challenging for phonological feature discrimination.

In out-of-domain scenarios, where both phoneme and phonological feature models are trained solely on native data, the impact of phonological features becomes more pronounced. We repeated the experiment using models trained on the TIMIT and LS training sets. Out of 34 confused phoneme pairs, 30 and 32 had at least one phonological feature with a FAR lower than the phoneme model when trained on TIMIT and LS, respectively. The average FAR improvement was 33%±37 for TIMIT and 30.5% ± 21.2 for LS models. Additionally, the FRR of phonological-level MDD was lower than that of phonetic-level MDD, with improvements of 22.7%±34, 30.4%±43.4, and 40%±21.7 for models trained on TIMIT-L2, TIMIT, and LS, respectively.

Fig. 14 provides further analysis of phonological features to explore their discrimination abilities. For each discriminative phonological feature, we computed the ratio between occurrences where the underlying feature obtained a lower FAR than its phoneme-level counterpart and

Table 5
The phoneme-level and phonological-level FRR and FAR of the most **consonant** common replacement errors in the L2-arctic test set. FAR and FRR were obtained from an in-domain model trained on the TIMIT_L2 dataset.

Consonant					
Conf_ph	Phonetic		Phonological		
	FAR	FRR	Feature	FAR	FRR
d/dh	**18.36**	18.10	dental	19.69	**9.49**
s/z	**35.85**	6.81	voiced	45.28	**5.92**
d/t	81.20	5.60	voiced	**70.94**	**3.54**
b/p	78.65	**1.57**	voiced	**68.54**	2.10
f/v	78.79	**2.90**	voiced	**68.18**	4.08
l/w	11.32	2.84	alveolar	**9.43**	**2.40**
t/th	20.63	5.97	dental	**19.05**	**2.49**
g/k	86.11	**1.15**	voiced	**73.61**	2.09
ch/jh	**62.79**	5.65	voiced	69.77	**3.32**
n/ng	56.06	**1.93**	velar	**46.97**	2.46
v/w	60.00	2.84	velar	**36.00**	**2.59**
s/th	69.70	3.33	dental	**42.42**	**2.68**
s/t	5.13	3.59	stop	5.13	**2.58**
s/sh	**10.53**	1.91	anterior	21.05	**1.49**
b/v	38.71	3.06	fricative	**25.81**	**2.22**
dh/z	85.71	33.53	dental	**57.14**	**14.82**
m/n	80.95	**1.29**	bilabial	**66.67**	1.40

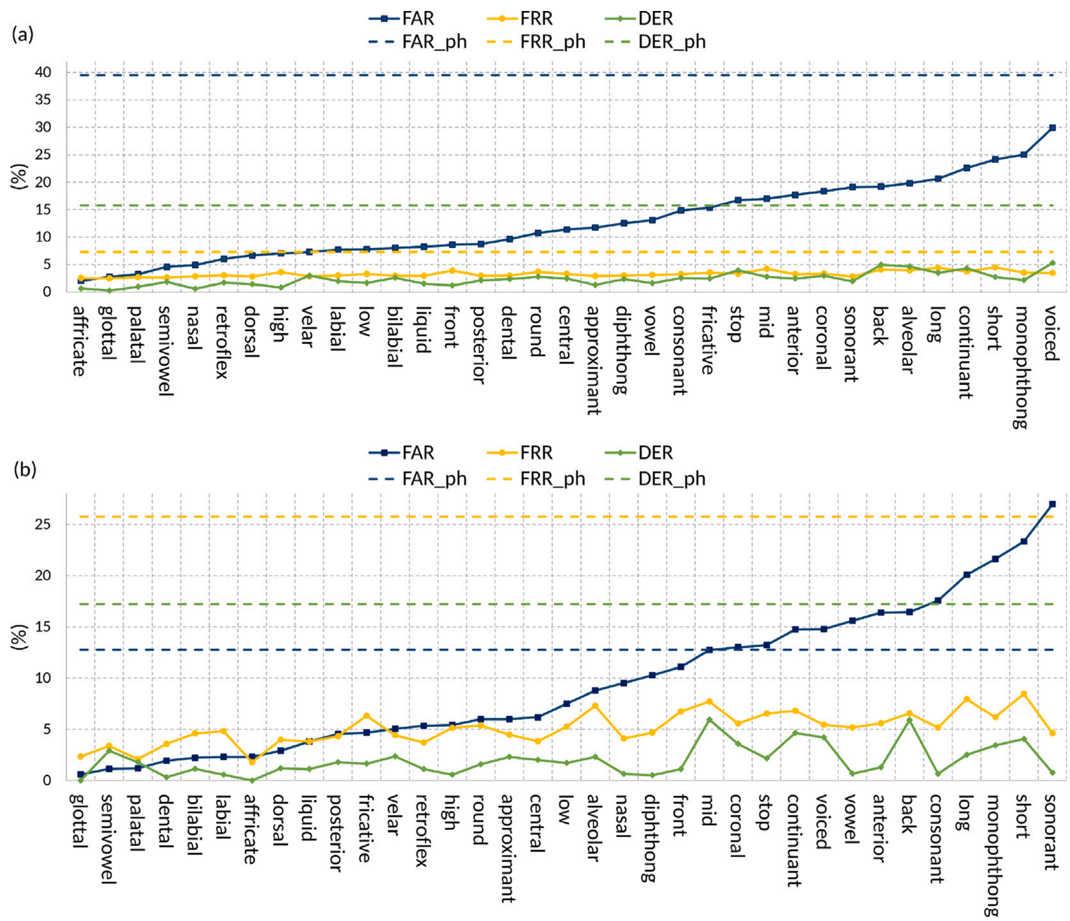

Fig. 12. The phonological-level MDD performance of (a) L2-Scripted and (b) L2-Suitcase. The phonological features are sorted from lowest to highest FAR. The straight lines FAR_ph, FRR_ph and DER_ph are the phoneme-level MDD metrics. For L2-scripted the FAR for all features detected by the phonological-level MDD was lower than the phoneme level MDD. The DER of the scripted test set (L2-Scripted) was consistently lower than that of the spontaneous test set (L2-Suitcase).

Table 6
The phoneme-level and phonological-level FRR and FAR of the most **vowel** common replacement errors in the L2-arctic test set. FAR and FRR were obtained from an in-domain model trained on the TIMIT_L2 dataset.

Vowels					
Conf_ph	Phonetic		Phonological		
	FAR	FRR	Feature	FAR	FRR
ih/iy	70.49	6.62	long	**65.57**	**5.07**
ao/ow	73.48	11.39	diphthong	**59.09**	**4.71**
ah/er	27.03	9.95	retroflex	**18.92**	**2.76**
eh/ey	81.67	**4.64**	diphthong	**51.67**	5.04
ah/ao	52.94	8.50	long	**45.10**	**5.96**
ae/eh	65.12	3.41	continuant	**55.81**	**2.88**
aa/ah	63.89	9.71	mid	**47.22**	**7.73**
ah/eh	63.89	7.49	front	**36.11**	**4.23**
aa/ae	41.94	7.56	front	**38.71**	**3.42**
aa/aw	68.00	12.48	central	**44.00**	**3.96**
aa/ao	44.00	**11.54**	round	**36.00**	13.66
ah/ih	80.00	8.54	back	**35.00**	**5.19**
ah/ow	52.63	9.67	round	**26.32**	**3.47**
ao/er	40.00	11.75	central	**13.33**	**9.36**
eh/ih	73.08	6.79	high	**46.15**	**4.04**
uh/uw	84.21	**6.28**	short	**47.37**	7.96
aa/ay	33.33	7.99	diphthong	**25.00**	**3.13**

the total number of occurrences where this feature appeared as a discriminative feature. In the in-domain scenario (TIMIT_L2), phonological features of vowels exhibit a more pronounced effect in distinguishing between confused phonemes compared to the out-of-domain scenario (TIMIT, LS). Conversely, consonant features such as *dental* and *fricative* consistently outperform the phoneme-level model in almost all cases in the out-of-domain scenario, while achieving lower performance in the in-domain scenario, with success rates as low as 40% and 20%, respectively.

Fig. 15 shows an example from the L2-Scripted data of the word ‘combination.’ The expected pronunciation of the /B/ sound was mispronounced by the speaker as /P/. The phoneme-based MDD model failed to detect this error. However, the phonological feature-based model successfully identified the mispronunciation by detecting the sound as unvoiced, which accurately corresponds to the unvoiced /P/ sound.

7. Conclusion

This paper introduced a novel MDD method to detect the existence of pronunciation errors and provide comprehensive diagnostic information. Unlike the current SOTA phoneme-level MDD which can only recognize the incorrect phoneme, our method, in addition to locating the mistaken phoneme, gives a detailed description of the pronunciation error in terms of which articulators are involved in the error’s production.

This was achieved by first modelling the phonological features, which include the manners and places of articulation. Given the recent superiority of the pre-trained speech representation model in different

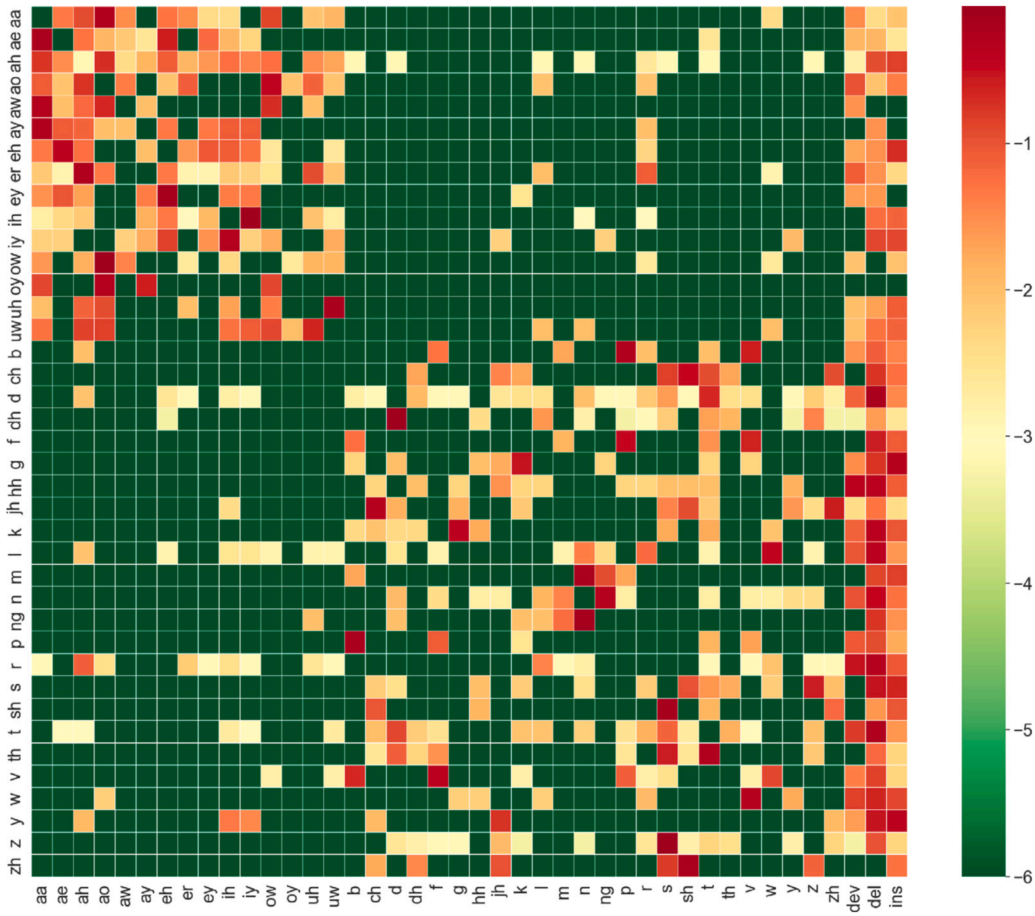

Fig. 13. Pronunciation errors made by L2 speakers of L2-ARCTIC speech corpora. The vertical axis labels show the canonical phoneme, while the horizontal axis labels show the replaced phoneme, in case of substitution error, the deviation (dev), insertion (ins), or deletion (del). The colour represents the percentage of each error relative to the total number of errors of each phoneme on a log scale with red indicating the highest confusion rate and green indicating no confusion. There is a notable occurrence of both deletion and insertion errors across all sounds, particularly consonants. Additionally, within the category of consonants, there is a significant level of confusion between voiced and unvoiced sounds, such as /s/ and /z/, /p/ and /b/, as well as /d/ and /t/. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

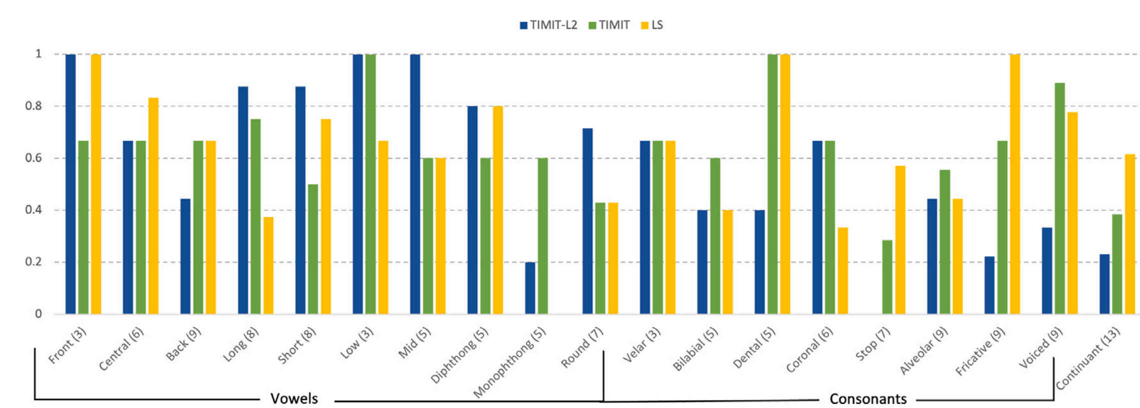

Fig. 14. The ratio between instances where each phonological feature achieved a lower False Acceptance Rate (FAR) compared to its phoneme-level counterpart and the total occurrences where it acted as a discriminative feature appears between brackets.

downstream tasks, we adopted the wav2vec2 pre-trained model as the core architecture of our phonological feature detection model. The results show that the large wav2vec2 model, which was pre-trained using data from different domains (clean, noisy, telephone, crowd-sourcing), outperformed single-domain models (see Fig. 6).

We further proposed a novel multi-label variant of the CTC loss function (SCTC-SB) to handle the non-mutually exclusive nature of phonological features. This enables the use of a single network for the joint modelling of all phonological features, making the model efficient in speed and memory. This is in contrast to current methods which require separate models for each feature (Lee et al., 2007; King and Taylor, 2000) or a group of mutually exclusive phonological features (Abraham et al., 2016; Merkx and Scharenborg, 2018). Furthermore, the results demonstrate the superiority of the proposed phonological feature detection model over End2End Deep Speech 2-based models (Amodei et al., 2016) for the detection of nasality (Cernak

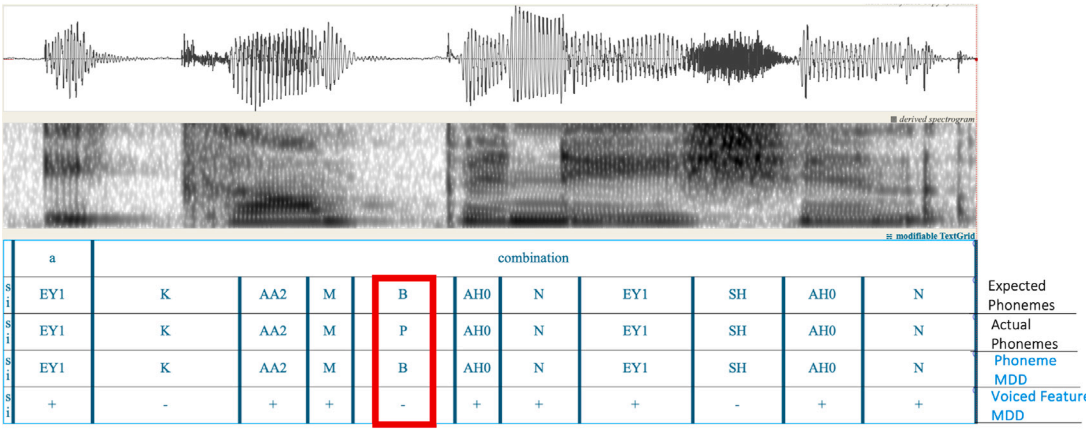

Fig. 15. The phonological feature-based model accurately detected the mispronunciation of the /B/ sound as unvoiced, aligning with the speaker's actual pronunciation of /P/, which the phoneme-based MDD failed to identify.

and Tong, 2018) and manner of articulations (Pradeep, 2018). (See Section 6.1.3)

The resultant phonological feature detection model was then used in performing MDD on non-native English speech corpus collected from 24 speakers of 6 different native languages. The experimental results show that our phonological-level MDD can achieve a reasonable performance (with FAR below 30% and DER below 10%) when trained solely on native speech corpora and applied to non-native speech (see Fig. 11). On the other hand, phoneme-level MDD which was trained and tested on the same native and non-native datasets achieved FAR and DER of 57% and 31% respectively (see Table 4). This is a significant advantage of the system as one of the major impediments in developing an accurate MDD is the scarcity of the non-native training dataset.

Furthermore, adding 2 h of annotated non-native data with 4 h of native data achieved an average correct detection rate of 88% and an average diagnosis accuracy of 97% in the phonological MDD (see Fig. 11).

Moreover, for certain frequently occurring substitution error pairs, employing a single phonological feature yields higher discrimination accuracy compared with phoneme models. For instance, utilizing the voicing feature proves more effective in distinguishing between the sounds /d/ and /t/ than their phonetic acoustic models. In addition to a higher accuracy compared to phoneme-based MDD, phonological-level MDD provides a low-level description of the pronunciation errors that is directly related to the articulatory system allowing formative feedback to be constructed.

Our future work includes leveraging the universal nature of the phonological features and incorporating speech corpora from multiple languages to train the phonological feature detection model. This will add more variations to the model leading to more robust MDD system. Furthermore, we are planning to extend the MDD system to more challenging domains such as adult and child disordered speech.

CRediT authorship contribution statement

Mostafa Shahin: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Julien Epps:** Supervision. **Beena Ahmed:** Conceptualization, Funding acquisition, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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